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AADK TASK 3

Task 3: 30-Day Compose Roadmap & Realization Plan

MindMatrix

30-Day Project Vision

Project Name: Compose Heroes Hub

Over 30 days, I will build a complete Superheroes app using Jetpack Compose. The app will include a themed heroes list, search & filter, favourites, detail screens, animations, accessibility support, and automated UI tests. The goal is to integrate all the concepts learned into one mini product.

Milestone Planning

Milestone 1 (Day 1–6): Project Setup & Foundation

Goal: Establish strong project base.

Tasks:

- Create new Compose project
- Setup Material 3 theming (light/dark mode)
- Define Superhero data model
- Create sample dataset (10–20 heroes)
- Setup basic navigation structure

Expected Outcome:

- App runs with basic scaffold and theme
- Clean folder structure (ui, data, viewmodel)

Success Criteria:

- Dark mode works
- Navigation compiles without errors
- Dataset displays in log or preview

Milestone 2 (Day 7–12): Heroes List UI

Goal: Build main list screen.

Tasks:

- Create Heroes List Screen
- Implement LazyColumn
- Design HeroCard UI
- Load images using Coil
- Add proper spacing & theme tokens

Expected Outcome:

- Scrollable list with 20 heroes
- Clean card UI with image, name, description

Success Criteria:

- Smooth scrolling
- No layout overflow
- Images load properly

Milestone 3 (Day 13–18): Search, Filter & Favourites

Goal: Add interaction logic.

Tasks:

- Implement search bar
- Add filter (Show Favourites)
- Implement favourite toggle logic
- Manage state using ViewModel
- Preserve state on rotation

Expected Outcome:

- Search filters heroes dynamically
- Favourites update instantly

Success Criteria:

- Search works without lag
- Favourite state persists after rotation

Milestone 4 (Day 19–24): Detail Screen & Animations

Goal: Add navigation and polish.

Tasks:

- Create Hero Detail Screen
- Pass selected hero via navigation
- Add simple animations (fade/expand)
- Add Snackbar feedback
- Improve UI polish

Expected Outcome:

- Navigation between list & detail works
- Animations smooth and meaningful

Success Criteria:

- No crashes during navigation
- Animations do not cause jank

Milestone 5 (Day 25–30): Accessibility, Testing & Performance

Goal: Final refinement & quality assurance.

Tasks:

- Accessibility audit (content descriptions, touch size)
- Add Compose UI tests (minimum 5)
- Performance testing (large dataset)
- Image optimization
- Prepare final presentation/demo

Expected Outcome:

- App accessible with TalkBack
- All UI tests passing

Success Criteria:

- No accessibility errors
- Tests run successfully
- Smooth performance with 50+ heroes

Resource & Tooling Plan

Tools & Libraries:

- Jetpack Compose (Material 3)
- ViewModel + Kotlin Flow
- Coil (image loading)
- Navigation Compose
- Accompanist (placeholders)
- Android Studio Hedgehog+
- GitHub (version control)
- GitHub Actions (CI for tests)

Learning Resources (Weekly)

Week 1:

- Official Compose documentation
- Material 3 theming guide
- Navigation Compose docs

Week 2:

- Compose LazyColumn best practices
- Coil documentation
- State management in Compose (Google codelabs)

Week 3:

- Compose animation documentation
- ViewModel + Flow tutorials
- Performance optimization blogs

Week 4:

- Accessibility in Compose guide
- Compose UI testing docs
- WCAG accessibility basics

Risk & Mitigation Table

Risk	Impact	Mitigation Strategy
Large dataset slows UI	Laggy scrolling	Use LazyColumn, stable keys, pagination
Animations cause jank	Poor UX on older devices	Test on mid-tier device, reduce animation duration
Accessibility test failures	App not inclusive	Weekly accessibility audit
State loss on rotation	Poor user experience	Use ViewModel + rememberSaveable
Time management issues	Delay in completion	Follow milestone deadlines strictly

Reflection & Commitment

I am excited about this 30-day project because it feels like building a real product, not just small practice screens. It integrates UI design, state management, animations, accessibility, and testing together. I expect to improve my planning skills and confidence in building scalable Compose apps. I will measure progress milestone-by-milestone and push code to GitHub every 2–3 days to stay accountable. Completing this roadmap will help me feel industry-ready.

Edge Test Cases

Test Case 1

Theme changes while card expansion animation is in progress.
Expected: UI should smoothly adapt colors without crashing or resetting state.

Test Case 2

User rapidly toggles favourite on a slow device.
Expected: Animation should remain smooth and not stack or freeze.

6. Reflection

The biggest challenge was balancing theme consistency with animation smoothness. I had to carefully ensure that animated values still used theme tokens instead of hardcoded values.
Combining theme and animation makes the UI feel alive and branded at the same time. Theme provides consistency, while animation provides interaction feedback.
However, developers should avoid adding animation everywhere. Too much motion can reduce clarity and impact performance. Animation should support functionality, not distract from it.